

World-in-World Genesis: Local Comparison Geometry and a Globalization Program

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Abstract

We give the precise mathematical content of the world-in-world proposal and separate it from its physical interpretation. The central object is a comparison field $Q_{\text{WW}} \in \Gamma(\text{Hom}(TP, TI))$ between two oriented rank-three bundles. Its nine local components split, at a nonsingular background, into three orientation and six strain directions. A selected flag resolves the strain sector into scalar, diagonal-shape, and shear sectors of dimensions $1 + 2 + 3$. Therefore $1 + 3 \times 3 = (1 + 3) + (1 + 2 + 3) = 4 + 6$ is an exact component identity. It is not ordinary manifold-dimension multiplication and does not derive a ten-dimensional spacetime, Lorentzian signature, gravity, matter, or quantization. We state the missing globalization theorem to the selected q79 Fu–Yau carrier, distinguish the shared phase circle from physical time, and formulate the additional gates needed for each proposed physical emergence. The resulting paper is a rigorous local geometry theorem plus a falsifiable research program.

Revision note for this edition

Supersedes. *World-in-World Genesis: A Proto-Geometric Origin of Time, Gravity, Matter, and Quantization*, version 4.

Reason. A three-dimensional base and three-dimensional fiber were multiplied into nine manifold dimensions, and the resulting count was used to claim automatic emergence of spacetime and several physical sectors.

Resolution. Version 5 replaces that construction by $Q_{\text{WW}} \in \Gamma(\text{Hom}(TP, TI))$, proves the local polar and $1+2+3$ strain decompositions, and states separate globalization/physical gates.

Retained result. The exact component identity $1 + 3 \times 3 = (1 + 3) + (1 + 2 + 3) = 4 + 6$ survives as local comparison geometry.

Remaining boundary. The q79 bundle-and-connection intertwiner, Lorentzian completion, gravity, matter, and quantization are not yet generated by this identity.

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1 How to Read World-in-World Geometry

The closest standard mathematical picture is the deformation gradient in continuum mechanics. A three-dimensional body can carry a 3×3 linear map comparing two local frames without becoming nine-dimensional. The nine numbers describe how one three-dimensional fiber is compared with another at the same base point. Polar decomposition then separates a local rotation from a positive strain.

World-in-world geometry uses this comparison structure as a candidate pre-projection language. The local linear algebra is exact. Its proposed relation to a four-dimensional Lorentzian base and six-dimensional q79 internal carrier is a separate globalization problem.

1.1 A concrete strain split

For example, consider the symmetric matrix

$$S = \begin{pmatrix} 2 & 4 & 5 \\ 4 & -1 & 6 \\ 5 & 6 & 2 \end{pmatrix}.$$

Its trace is 3, so the scalar part is

$$P_1 S = I_3.$$

Relative to the displayed ordered frame, the traceless diagonal and off-diagonal parts are

$$P_2 S = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad P_3 S = \begin{pmatrix} 0 & 4 & 5 \\ 4 & 0 & 6 \\ 5 & 6 & 0 \end{pmatrix}.$$

They add back to S and are mutually orthogonal in the Frobenius inner product. The spaces from which these pieces are drawn have dimensions 1, 2, and 3. The numerical entries are not those dimensions, and the three subspaces are not three separate manifolds.

1.2 The logical pipeline

The paper advances through four stages:

1. define one local comparison field and prove its orientation/strain decomposition;
2. select a flag and obtain the local $1 + 2 + 3$ strain profile;
3. compare that profile with the independently selected q79 rank-1 + 2 + 3 carrier; and
4. require an isometric, connection- and operator-preserving intertwiner before treating the two carriers as one physical construction.

Only the first two stages are local consequences of Q_{WW} . The third provides a nontrivial structural clue. The fourth is the proposed globalization theorem and remains the decisive bridge.

2 Claim discipline

This paper has three layers:

Proved local geometry. Bundle dimensions, polar decomposition, and the orthogonal component splittings stated below.

Selected downstream data. The q79 trace-split carrier and Fu–Yau branch as recorded by their independent theorem packets.

Research program. The proposed identification of local strain sectors with q79 vertical operators and, after that, with physical spacetime fields.

No statement in the third layer is promoted by suggestive equality of dimensions. A bridge must identify maps, transition functions, connections, actions, and observables.

3 The local comparison field

Let B be a smooth manifold and let $TP, TI \rightarrow B$ be oriented Euclidean rank-three bundles. The labels “outer” and “inner” describe the roles of the two fibers; they do not create separate universes or coordinate factors.

Definition 3.1 (World-in-world field). A world-in-world field is

$$Q_{\text{WW}} \in \Gamma(\text{Hom}(TP, TI)).$$

Under changes of oriented orthonormal frames $g_P, g_I : B \rightarrow SO(3)$, its local matrix transforms as

$$Q_{\text{WW}} \longmapsto g_I Q_{\text{WW}} g_P^{-1}.$$

This transformation law is essential: the nine entries are not nine scalar fields independent of frame choice.

Theorem 3.2 (Bundle dimension versus comparison components). *If $\dim B = 3$, the total space of either rank-three vector bundle has dimension six. The bundle $\text{Hom}(TP, TI)$ has rank nine, so a section has nine local fiber components.*

Proof. Local bundle charts have the forms $U \times \mathbb{R}^3$ and $U \times \text{Mat}(3, \mathbb{R})$, respectively. Dimensions are $3 + 3 = 6$ and ranks are $3 \cdot 3 = 9$. $\square$

Thus the original intuition survives only as “three-dimensional outer space plus a 3×3 comparison field.” It does not survive as three independent three-dimensional manifolds.

4 Polar and strain decomposition

Assume $Q_{\text{WW}}(b) \in GL^+(3, \mathbb{R})$ on an admissible domain. Polar decomposition is unique:

$$Q_{\text{WW}} = RU, \quad R \in SO(3), \quad U = (Q_{\text{WW}}^T Q_{\text{WW}})^{1/2} > 0.$$

The logarithmic strain is

$$S = \log U \in \text{Sym}(3, \mathbb{R}).$$

Theorem 4.1 (Orientation–strain normal form). *At every nonsingular background, the tangent space of the comparison carrier has the orthogonal Frobenius decomposition*

$$\text{Mat}(3, \mathbb{R}) = \mathfrak{so}(3) \oplus \text{Sym}(3, \mathbb{R}), \quad 9 = 3 + 6.$$

Proof. Every matrix X decomposes uniquely as $X = (X - X^T)/2 + (X + X^T)/2$. Skew and symmetric matrices are orthogonal for $\langle X, Y \rangle_F = \text{tr}(X^T Y)$ and have dimensions three and six. $\square$

Choose an orthonormal flag, equivalently a preferred ordered frame modulo the declared residual symmetry. For $S \in \text{Sym}(3)$ define

$$P_1 S = \frac{\text{tr } S}{3} I_3, \quad P_2 S = \text{diag}(S) - P_1 S, \quad P_3 S = S - \text{diag}(S).$$

Theorem 4.2 (Selected $1 + 2 + 3$ strain split). *The maps P_1, P_2, P_3 are mutually orthogonal projectors with ranks 1, 2, 3, and*

$$S = P_1 S + P_2 S + P_3 S, \quad \|S\|_F^2 = \sum_{j=1}^3 \|P_j S\|_F^2.$$

Proof. $P_1 S$ is scalar diagonal, $P_2 S$ is traceless diagonal, and $P_3 S$ has zero diagonal. These subspaces are pairwise Frobenius-orthogonal, their dimensions sum to six, and the displayed formulas follow directly. $\square$

The $1 + 2 + 3$ split is flag-dependent. Without a selected flag, the canonical $SO(3)$ -equivariant split is $1 + 5$ into trace and traceless symmetric tensors. The flag is therefore source data that must either be selected by q79 geometry or declared as an assumption.

5 What the ten-component identity says

If one additional scalar u records an ordering coordinate, the local data have the count

$$\underbrace{1}_u + \underbrace{9}_{Q_{\text{WW}}} = \underbrace{(1+3)}_{u \text{ and orientation}} + \underbrace{(1+2+3)}_{\text{strain}} = 4 + 6.$$

Proposition 5.1 (No automatic spacetime theorem). *The preceding equality does not imply a splitting $TM_{10} \cong TY_4 \oplus TX_6$, does not select a Lorentzian metric on a four-dimensional factor, and does not identify the strain bundle with a compact internal tangent bundle.*

Proof. A component count contains no transition functions for TM_{10} , no integrable distributions, no metric of Lorentzian inertia, and no compactness or topology data. Each is independent structure. $\square$

The identity is nevertheless useful: it gives a precise local carrier whose four orientation/order and six strain components can be compared with a separately supplied $4 + 6$ physical realization.

In other words, the count is a compatibility test for a proposed map, not a construction of that map. A mismatch would rule out a simple component-preserving identification. A match allows the comparison to proceed to topology, connection, and operator tests, where genuinely global obstructions can appear.

6 Shared circle and physical time

Let $L_{\text{shared}} \rightarrow X_6$ be a Hermitian line bundle with unitary connection. Its circle fibers encode common phase or holonomy. Reusing this line in all three rank lanes does not add three circle coordinates and does not create a seventh internal dimension.

Physical time in the canonical completion is a noncompact ordering variable on a Lorentzian base, represented by a propagator $U(t_2, t_1)$. A phase map

$$\mathbb{R} \longrightarrow S^1, \quad t \longmapsto e^{i\omega t},$$

can synchronize phase with time after t and ω are supplied, but it is many-to-one and cannot make compact phase identical to physical time. An arrow of time further requires an asymmetric dynamical law or boundary condition.

7 Selected q79 global carrier

For the selected degree-three cover $\pi : C \rightarrow B$, define

$$\mathcal{A} = \pi_* \mathcal{O}_C, \quad \mathcal{A}_0 = \ker \text{Tr}, \quad \mathcal{A} = \mathcal{O} \oplus \mathcal{A}_0.$$

The selected common carrier is

$$\mathcal{H}_{\text{q79}} = L_{\text{shared}} \otimes (\mathcal{O} \oplus \mathcal{A}_0 \oplus \mathcal{A}), \quad \text{rank} = 1 + 2 + 3 = 6.$$

The trace split is global and intrinsic to the cover. It is not a global ordered-sheet flag: connected transitive monodromy prevents such an ordering. The signed sheet representation has a local binary-dihedral spin lift, but strict global Spin closure still requires all branch-complement relator signs or the equivalent obstruction class.

The q79 Fu–Yau branch is the current selected compactification candidate. The auxiliary space $L(3, 1) \times \text{Nil}_3$ is not homeomorphic to it and is used only as an effective model or operator mnemonic. Literal $S^1 \times \text{Lens} \times \text{Nil}$ and literal manifold nesting are not premises of this paper.

8 The globalization theorem

The central open map is a bundle morphism

$$\mathfrak{J} : \mathbb{R}I_3 \oplus \mathcal{D}_0 \oplus \mathcal{O} \longrightarrow L_{\text{shared}} \otimes (\mathcal{O} \oplus \mathcal{A}_0 \oplus \mathcal{A}).$$

Conjecture 8.1 (Selected world-in-world/q79 intertwiner). On the selected q79 Fu–Yau branch there is a source-selected flag and an isometric bundle isomorphism $\mathfrak{I}$ preserving the $1 + 2 + 3$ filtration such that

$$\mathfrak{I}\nabla^{\text{WW}} = \nabla^{\text{HYM}}\mathfrak{I}, \quad \mathfrak{I}H_{\text{WW}} = H_{\text{q79}}\mathfrak{I},$$

and likewise for the retarded and overlap kernels used in the numerical source packets.

The conjecture has four independent tests:

1. transition functions agree on all overlaps;
2. the selected flag is monodromy-compatible rather than an ordered-sheet choice;
3. the two inner products and connections are intertwined; and
4. the resulting operator coefficients equal the source coefficients used downstream, with a declared normalization and error certificate.

Rank matching proves none of these clauses, but it makes the conjecture well-typed and finite.

The conjecture is therefore falsifiable in several independent ways. The bundles may have incompatible characteristic data, monodromy may obstruct the flag, the connections may fail to be conjugate, or the transported Hessian may disagree with the downstream source operator. Failure of any one test would end this proposed identification without undoing the local orientation/strain theorems.

9 Physical emergence gates

The original narrative can now be expressed as separate targets.

9.1 Time

A noncompact parameter, physical propagator, and arrow-producing condition are required. Bookkeeping order or phase winding alone is insufficient.

9.2 Gravity

A Lorentzian metric or tetrad, connection, action or equations, constraints, and a stress tensor are required. The local polar factor R does not derive the Einstein–Hilbert action.

9.3 Matter

A spin bundle, finite representation data, Dirac operator, chirality and real structure, and a particle/observable interpretation are required. The local $1 + 2 + 3$ strain split alone does not select Standard Model representations.

9.4 Quantization

A complex state space, observable algebra, dynamics, probability functional, and gauge and constraint treatment are required. Discrete topology or a compact resolvent can produce discrete spectra but not the Born rule by itself.

9.5 Mass and scalar relaxation

Mass requires a canonically normalized pole or dispersion relation. A Higgs mode requires a selected scalar representation and couplings. Positivity of a six-dimensional strain Hessian does not imply a unique radial Higgs.

10 Relation to current calculations

The finite 27×27 matrix, charged Yukawa profile, CKM, electroweak, threshold, and precision packets close embedded renormalized-SM equivalence at the declared one-shared-physical-primitive/profile standard. They constitute a nontrivial downstream realization. Standard BRST/Faddeev–Popov quantization and measured renormalized profile data are admitted at that standard; strict no-knob selection remains an upgrade. These computations do not retroactively prove the world-in-world globalization conjecture.

11 Conclusion

World-in-world geometry has a clean surviving core. It is a rank-three comparison field with an exact orientation/strain decomposition and a selected $1 + 2 + 3$ local normal form. Its physical promise is concentrated into one testable globalization problem rather than many automatic-emergence claims. Failure to construct the q79 intertwiner would falsify the proposed unified identification while leaving the local linear algebra intact.

Computational Evidence and Reproducibility

The q=79, finite-matrix, HYM, and internal TT rows give concrete targets for the proposed globalization map. They do not prove the missing intertwiner from local comparison strain to the physical q=79 vertical geometry. Standard Model profile rows are context, and the strict upgrade remains open.

The referenced rows are frozen to the curated results repository state identified below. Hashes are grouped in eight-character blocks for line breaking.

Repository: <https://github.com/PeterNero/mtt-results-repro>

Commit: 31247ebb 5c22f3fb b5443024 365433c6 ee0bff4a

Manifest: release/result_manifest.json

Manifest SHA-256:

fb399689 60b00584 631dbf53 1a708e18

ef928d6b 6d935119 c185d7f6 32b1e7cd

Tier labels are quoted verbatim from that manifest. A row used directly supports only the specific computational statement identified above; a corpus-state cross-check does not prove this paper’s local theorems; and an open row is evidence of an unresolved obligation, never of closure.

Rows used directly in this paper.

- A13/gr_tt_support (*derived exact*).
Exact-branch internal TT support certificate; physical normalization remains open.
- A19/hym_wiener_contraction (*numeric certified*).
Weighted-theta Fourier-tail and Wiener contraction certificate.
- A11/q79_exact_audit (*derived exact*).
Executable q=79 exact-branch audit.
- A11/q79_exact_theorem (*derived exact*).
CRT q=79 theorem on the selected exact branch.
- A01/qutrit_weyl_27_matrix (*derived exact*).
Sparse 27x27 qutrit-Weyl left-action realization.

Corpus-state cross-checks.

- A01/charged_yukawa_higgs_profile (*profile replay*).
Versioned Yu, Yd, Ye and lambda_H profile packet.
- A14/ckm_prediction_profile (*numeric certified*).
Three selected CKM profile rows and uncertainty comparison.
- A01/current_global_lock (*profile replay*).
Current non-looping global status and source-certificate map.
- A01/direct_k_higgs_row (*derived exact*).
Promoted direct K_threshold.Omega_H.lambda row.
- A22/e6_qpsi_qcd_anomaly (*derived exact*).
E6 Qpsi matter/exotic QCD anomaly cancellation audit.
- A04/final_12_of_12_audit (*profile replay*).
Twelve-obligation embedded renormalized-SM equivalence audit.
- A40/neutral_two_primitive_profile (*profile replay*).
Measured two-splitting neutrino profile, masses and Dirac Yukawa rows.
- A02/precision_15_source_transport (*profile replay*).
Fifteen measured source coordinates, Jacobian and covariance transport.
- A02/precision_8x8_workspace (*profile replay*).
Eight-coordinate SMDR output with positive-definite 8x8 covariance.
- A01/strict_pew_row (*derived exact*).
Promoted P_EW source row at the declared one-shared-primitive standard.

Open boundary (not evidence of closure).

- A05/strict_upgrade_ledger (*open*).
Current 2/9 strict no-knob upgrade ledger.

No imported row changes theorem ownership or promotes a neighboring claim: all local statements retain their stated hypotheses, domains, and limitations.

References

- [1] S. Helgason, *Differential Geometry, Lie Groups, and Symmetric Spaces*, Academic Press, 1978.

- [2] J.-X. Fu and S.-T. Yau, *The theory of superstring with flux on non-Kähler manifolds and the complex Monge–Ampère equation*, J. Differential Geom. 78 (2008).
- [3] P. Nero, *MTT Foundational Geometry Reconciliation*, internal theorem and verification packet, 2026.